

## CONCLUDING REMARKS

Of these results and predictions (1), (2) and (3) are substantially identical with the conclusions put forward in 1921 in my paper, "Diffusion by Continuous Movements," where the suggestion that the diffusive power of turbulence should be used for the purpose of measuring the scale of turbulence and the turbulent components was first made. Recently experiments of this nature have been made by C. B. Schubauer\* and (2) has been verified, as will be shown in Part IV of the present paper. Mr. Schubauer, however, worked quite independently of my previous work and indeed gives an empirical explanation of his experimental results. Conclusions (4) to (9) are, I believe, new. It will be shown in Part II that all these results, except (9), have now been verified experimentally and shown to be true. Experimental work is now in hand to test the truth of (9).

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## Statistical Theory of Turbulence—II

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### MEASUREMENTS OF CORRELATION IN THE EULERIAN REPRESENTATION OF TURBULENT FLOW

The methods described in Part I have been used by Mr. L. F. G. Simmons, of the National Physical Laboratory, to find experimentally the correlation between the turbulent components of velocity  $u_0$  and  $u_y$  at two points distant  $y$  apart in a direction transverse to the stream. The measurements were made at mean speed  $U = 25$  feet per second in a wind tunnel behind a honeycomb with 0.9-inch square mesh. The results are shown in fig. 1 where the ordinates are  $R_y = \frac{u_0 u_y}{u^2}$  and the abscissae are the corresponding values of  $y$ . It will be seen that the  $R_y$  curve is apparently rounded at the top and that  $R_y$  falls to 0.08 at  $y = 0.38$  inches. No measurements were made beyond this point, but extrapolation seems to show that  $R_y = 0$  when  $y$  is about 0.5 inches, *i.e.*, when  $y$  is slightly greater than  $\frac{1}{2}M$ .

\* 'Rep. Nat. Adv. Ctee. Aero., Wash.,' No. 524 (1935).

Integrating the curve of fig. 1 the value of  $l_2 = \int_0^{0.5''} R_y dy = 0.175$  inch, so that

$$l_2/M = 0.175/0.9 = 0.195. \quad (1)$$

Another set of measurements of similar nature made in a parallel-wall tunnel 24.6 cm high by Prandtl and Reichardt is shown in fig. 1 of Part III, which is reproduced from their paper. Their method was not so suitable for measuring low values of  $R_y$  as that of Simmons, and their curves are not given for values of  $R_y$  less than 0.6.

The forms of the  $R_y$  curves near  $y = 0$  will be discussed later.

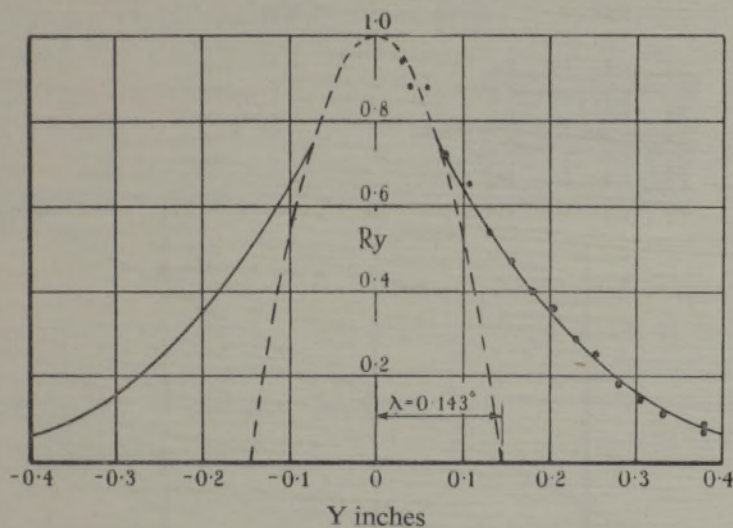


FIG. 1—Measured values of  $R_y = \frac{\overline{u_0 u_y}}{\overline{u_2^2}}$  behind 0.9 inch by 0.9 inch honeycomb;  $\frac{\overline{u^2}}{u^2} = 0.1015$  (ft sec)<sup>2</sup>

### DECAY OF TURBULENCE IN AN AIR STREAM BEHIND A GRID OR HONEYCOMB

Few observations of the decay of turbulence seem to have been made, and in most of them where observations from which the decay can be found have been made the experiment was concerned with other phenomena. In some such cases the necessary data for comparing with the theory developed in Part I of this paper were not all recorded.

#### SIMMONS AND SALTER'S MEASUREMENTS BEHIND GRIDS

The measurements contained in Table I were made by Simmons and Salter\* by means of hot wires in the stream behind a grid consisting of

\* 'Proc. Roy. Soc.,' A, vol. 145, p. 233 (see Table II, p. 233).



strips 0.5 inch wide forming square-mesh grid with 1-inch square holes so that  $M = 1.5$  inches = 3.81 cm.

TABLE I— $M = 1.5$  INCHES

| U                | x      | $u'^2$                 | $u'$   | $U/u'$ | $Mu'/\nu$ |
|------------------|--------|------------------------|--------|--------|-----------|
| ft sec           | inches | (ft sec) <sup>2</sup>  | ft sec |        |           |
| 5.19 }<br>7.2 }  | 6      | { 1.23 }<br>{ 2.02 }   | 1.27   | 4.88   | 1030      |
| 5.65 }<br>5.76 } | 9      | { 0.565 }<br>{ 0.614 } | 0.768  | 7.42   | 636       |
| 5.29             | 15     | 0.162                  | 0.402  | 13.15  | 332       |

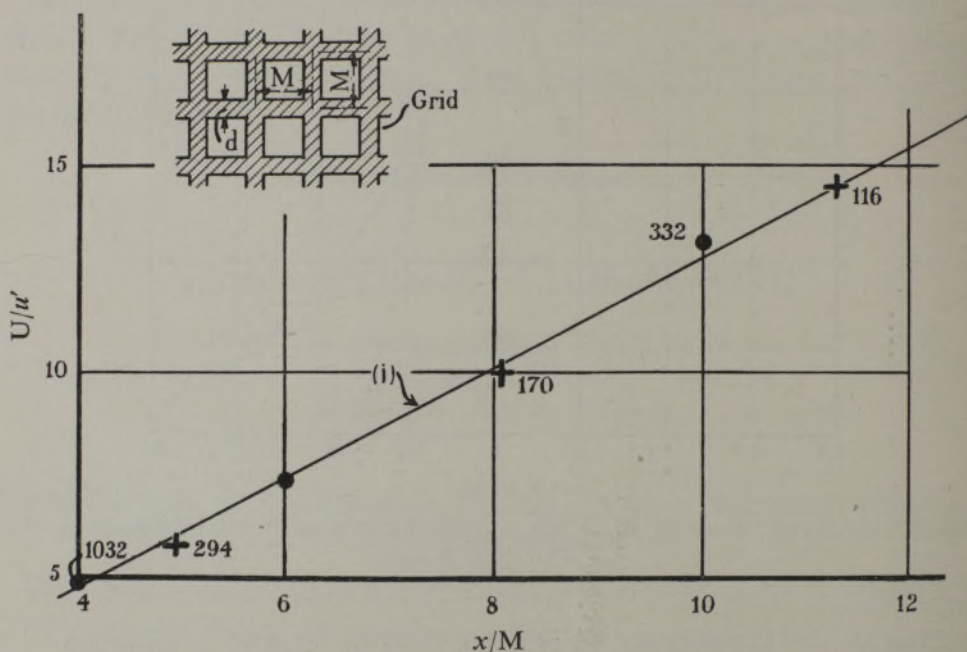


FIG. 2—Decay turbulence behind grid of flat strips; ● = 1.5 inches;  $d = 0.5$  inches; +  $M = 0.62$  inches; (i)  $U/u' = -0.7 + 1.32 x/M$

In this table  $u' = \sqrt{u'^2}$  and  $x$  is the distance behind the grid. At  $x = 6$  and 9 inches the "wind shadow" had not disappeared, the two sets of readings being those obtained behind the middle of a strip and behind the middle of a hole respectively. The values of  $u'$  and  $U/u'$  in columns 4 and 5 are deduced from means of the two sets of figures in columns 1 and 3.

According to the theoretical prediction (see equation (59), Part I)  $U/u'$  should increase linearly with  $x$ . Fig. 2 shows  $U/u'$  plotted against  $x/M$ . It will be seen that the prediction is very nearly verified, the points lying very close to a straight line.

The value of the constant  $A$  which occurs in (59) can be calculated from any two of the three sets of measurements. Taking from Table I the measurements at  $x = 6$  and 9 inches I find

$$A^2 = 3.94 \text{ so that } A = 1.98,$$

and taking the measurements at  $x = 9$  and 15,

$$A^2 = 3.49 \text{ so that } A = 1.87.$$

The Reynolds number of the turbulence, namely,  $Mu'/\nu$ , corresponding with the three stations are given in the last column of Table I.

It is of interest to compare the rate of decay of turbulence observed by Simmons and Salter with that which would result if the complete turbulence pattern remained constant through the range of observation from  $x = 6$  inches to 15 inches, but the turbulent velocity decreased owing to viscosity. The turbulent velocity components in that case\* would be proportional to  $e^{-ct}$ , so that

$$\frac{u'_{15}}{u'_6} = \left( \frac{u'_9}{u'_6} \right)^3. \quad (2)$$

The 3 appears as exponent in (2) because the time taken by the stream to travel the distance from  $x = 6$  inches to 15 inches is three times as long as the time for  $x = 6$  inches to 9 inches.

On the other hand, equation (59), Part I, can be reduced in this case to

$$\frac{u'_{15}}{u'_6} = \frac{u'_9/u'_6}{3 - 2u'_9/u'_6}. \quad (3)$$

From Table I it will be seen that  $u'_9/u'_6 = 0.605$ , thus the supposition that turbulent motion decays by the direct action of viscosity without any increase in the diameter of the smallest eddies leads to the prediction that

$$u'_{15}/u'_6 = (0.605)^3 = 0.22. \quad (A)$$

The present theory leads to the prediction that

$$u'_{15}/u'_6 = \frac{0.605}{3.2(0.605)} = 0.338. \quad (B)$$

The observed value is

$$u'_{15}/u'_6 = \frac{0.402}{1.27} = 0.317. \quad (C)$$

Comparing these three values ((A), (B), (C)) it will be seen that Simmons and Salter's observations are in fairly good agreement with the

\* Taylor, 'Phil. Mag.', vol. 46, p. 671 (1923).



present theory but do not agree with theories of dissipation which do not allow for increasing size of the smallest eddies with decrease in  $Mu'/\nu$ .

Some further unpublished observations made by Mr. Simmons with a grid geometrically similar to his  $1\frac{1}{2}$ -inch grid, but of smaller mesh ( $M = 0.62$ ), are also shown in fig. 2. It will be seen that when plotted non-dimensionally (*i.e.*, using  $x/M$  as abscissae) the points lie very close to those which represent the experiments with the larger grid. Thus the prediction that the constant  $A$  of equation (59), Part I, is a constant for all similar grids is verified. The question whether  $A$  is a universal constant, as is predicted in Part I, for all square-mesh grids irrespective of the ratio of the width of the bars of the grid to the mesh remains open so far as these observations are concerned.

#### DRYDEN'S OBSERVATIONS BEHIND HONEYCOMBS

The first reliable observations on the decay of turbulence behind a honeycomb are contained in a report by Dryden.\* In that report the size of the honeycomb was not given, but Dr. Dryden has very kindly supplied me with the necessary data. The honeycomb was hexagonal, the distances between opposite faces being 3 inches. Taking  $M = 3.0$  inches the observed values are given in Table II.

TABLE II

|        |      |      |      |
|--------|------|------|------|
| $x/M$  | 21   | 38   | 58   |
| $U/u'$ | 43.5 | 62.5 | 83.3 |

If  $U = 20$  ft/sec  $Mu'/\nu$  lies between 1000 and 2000. Similar observations behind a honeycomb of 3-inch circular tubes arranged in hexagonal piling gave the results shown in Table III.

TABLE III

|        |      |    |     |
|--------|------|----|-----|
| $x/M$  | 21   | 38 | 58  |
| $U/u'$ | 62.5 | 77 | 100 |

These results being obtained with honeycombs of circular and hexagonal tubes are not strictly comparable with those obtained with honeycombs or grids arranged in a pattern of squares. The difference, however, may be expected to be slight, and they are included because, as will be seen in fig. 6 where they are shown in graphic form, they illustrate how accurately the theoretical linear law of increase in  $U/u'$  is borne out in practice.

\* 'Rep. Nat. Adv. Cttee. Aero. Wash.,' No. 342, *see also* Report No. 392.

N.P.L. OBSERVATIONS BEHIND GRIDS AND HONEYCOMBS

Recently Simmons and Salter have made some more measurements of  $U/u'$ , using a hot-wire compensating circuits and a thermo-junction milliammeter to measure  $\overline{u^2}$  in the manner explained in Part I. A set of observations made behind a grid of round bars 0.875 inch diameter arranged in a 3-inch square mesh\* is shown in fig. 3. In the work previously described the average value of  $U/u'$  taken over a long series of observations made at the same section of the tunnel were given. In

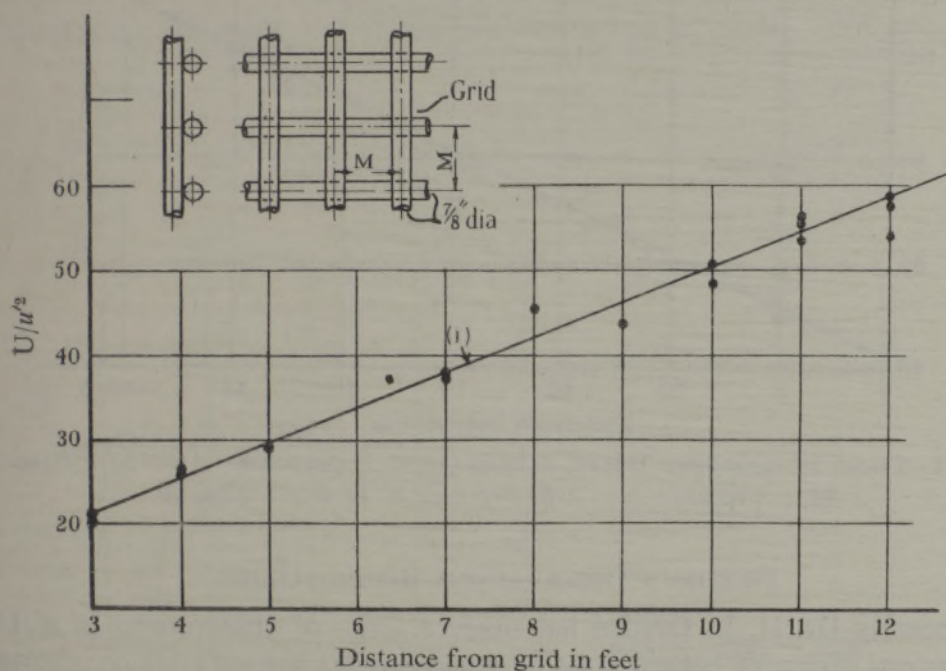


FIG. 3—Decay of turbulence behind a grid of round bars.  $U = 19.5$  ft/sec;  $M = 3$  inches;  $A = 2.20$ ; (i)  $U/u' = 8.9 + 1.035 x/M$

fig. 3 the actual observations or means of a small number of observations are shown. It will be seen that the linear law of increase in  $U/u'$  is in good agreement with the experiments. The line shown in the figure is  $\frac{U}{u'} = 8.9 + 1.035 \frac{x}{M}$ , which agrees with equation (59) of Part I if  $A = 2.20$ .

A similar set of measurements in a 4-foot wind tunnel behind a 3-inch square-mesh honeycomb is shown in fig. 4. Again it will be seen that the measurements confirm the theoretical linear law of increase in  $U/u'$ ,

\* As is indicated in fig. 3, this grid consisted of two sets of parallel rods in contact so that the central plane of one set is a distance  $d$  further down-stream than the other.



the line in this case being  $\frac{U}{u'} = 12.6 + 1.230 \frac{x}{M}$ , which agrees with (59), Part I, if  $A = 2.05$ .

Observations behind a 0.9-inch by 0.9-inch honeycomb in a 1-foot tunnel at 5 ft per sec gave a well-defined straight line

$$\frac{U}{u'} = 5.5 + 1.30 \frac{x}{M}. \quad (4)$$

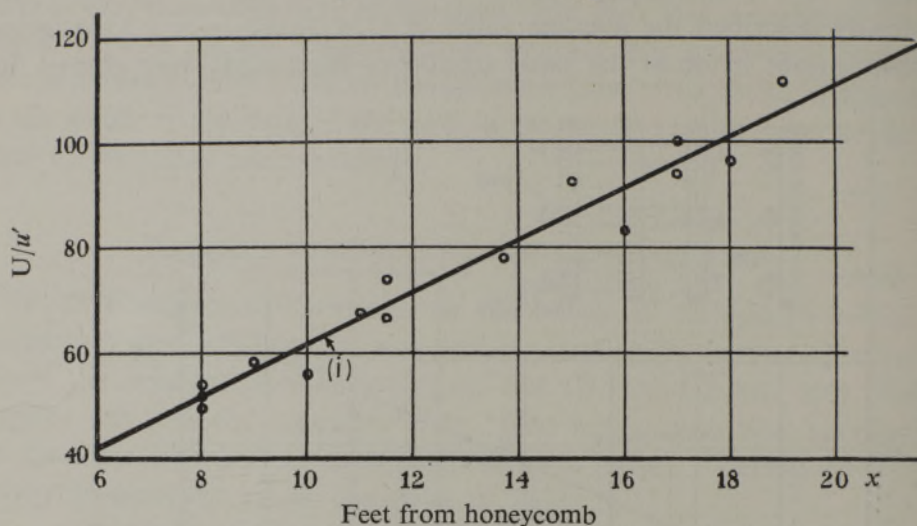


FIG. 4—Decay of turbulence behind a square-mesh honeycomb.  $U = 20.2$  ft/sec;  $M = 3$  inches;  $A = 2.05$ ; (i)  $U/u' = 12.6 + 1.23 x/M$

#### DRYDEN'S OBSERVATIONS BEHIND GRIDS

Recently Dr. H. L. Dryden has made a series of observations of  $u'/U$  at various distances behind a set of nearly similar square grids consisting of round bars each about 0.193 of the mesh length. The values of  $M$  were  $M = 5$  inches,  $3\frac{1}{4}$  inches, 1 inch,  $\frac{1}{2}$  inch,  $\frac{1}{4}$  inch,  $\frac{1}{8}$  inch. Dr. Dryden kindly set forth his results for me in the diagram shown in fig. 5 where the abscissae are  $x/d$ ,  $d$  being the diameter of the bars of the grid and  $x$  the distance down-stream from the grid. The numbers attached to the points are the mesh length  $M$  in inches. The straight line drawn in the diagram which passes nearly through the points corresponding with the  $M = 5$  inches and  $3\frac{1}{4}$  inches is

$$\frac{U}{u'} = 7 + 0.25 \frac{x}{d},$$

and since  $d = 0.193 M$  this is equivalent to

$$\frac{U}{u'} = 7 + 1.30 \frac{x}{M}. \quad (5)$$

This line fairly represents the observations up to  $U/u' = 100$ , but for higher values of  $U/u'$  the turbulence does not decrease so rapidly as the theoretical formula would lead one to expect.

The points on the extreme right of the diagram, fig. 5, correspond with very low values of  $Mu'/\nu$ , namely, 7 and 14. This is well below the lowest value of  $Mu'/\nu$  at which the linear law would be expected to hold, but, as has been pointed out, the rate of decay for a given scale of turbulence at low values of  $Mu'/\nu$  must be *greater* than that indicated by the linear law. Fig. 5 shows that in Dr. Dryden's tunnel the turbulence dies away more slowly than the linear law when the turbulence is small. This indicates

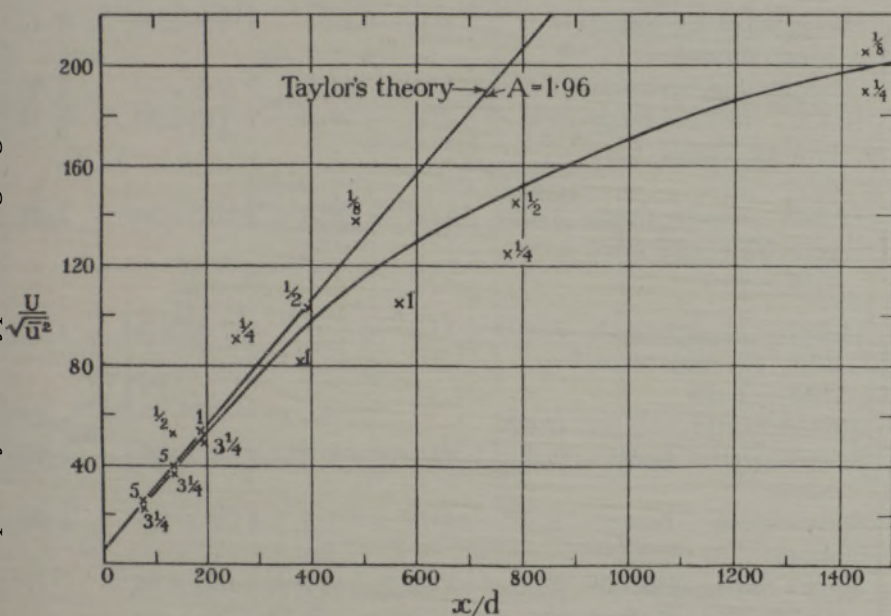


FIG. 5—Decay of turbulence behind a grid of round bars (H. L. Dryden)

that the large-scale turbulence which is in the air stream before striking the grid can pass through it. Near the grid this is masked by the more violent disturbances produced by the grid itself. As the distance downstream from the grid increases the large-scale turbulence which has passed through it decreases more slowly than that produced by the grid and eventually completely obliterates the effect of the grid.

It seems likely that a honeycomb with long cells might be more likely to reduce the pre-existing turbulence to a definite scale than a grid of round bars, and in fact the theoretical linear law of decay seems to be obeyed very consistently in the disturbances behind honeycombs.



## SUMMARY OF RESULTS ON DECAY OF TURBULENCE

All these results are collected together in fig. 6 and the equations to the straight lines which pass most nearly through the observed points are collected together in Table IV. Dr. Dryden's results with small-mesh honeycombs,  $M = 1$  inch,  $\frac{1}{2}$  inch,  $\frac{1}{4}$  inch,  $\frac{1}{8}$  inch, have not been included because the corresponding values of  $LU/\nu$  are so small that the theory

TABLE IV

| Authority                | —                                                         | M<br>inches | Equation to line in<br>fig. 6 | A    |
|--------------------------|-----------------------------------------------------------|-------------|-------------------------------|------|
| Dryden                   | Honeycomb of round<br>3-inch tubes                        | 3.0         | $U/u' = 37.1 + 1.056 x/M$     | 2.18 |
|                          | Hexagonal honeycomb                                       | 3.0         | $U/u' = 21.1 + 1.078 x/M$     | 2.16 |
|                          | 5-inch square-mesh<br>grid of round bars<br>$d/M = 0.2$   | 5.0         | $U/u' = 9.5 + 1.110 x/M$      | 2.12 |
|                          | 3½-inch square-mesh<br>grid of round bars<br>$d/M = 0.2$  | 3.25        | $U/u' = 8.8 + 1.072 x/M$      | 2.16 |
|                          | 1½-inch square-mesh<br>grid of flat slats<br>$d/M = 0.33$ | 1.50        | $U/u' = -0.7 + 1.32 x/M$      | 1.95 |
| Simmons<br>and<br>Salter | 0.62-inch grid of slats                                   | 0.62        |                               |      |
|                          | Square-mesh honey-<br>comb                                | 0.90        | $U/u' = 5.5 + 1.30 x/M$       | 1.96 |
|                          | Square-mesh honey-<br>comb                                | 3           | $U/u' = 12.6 + 1.23 x/M$      | 2.05 |
|                          | Square-mesh grid of<br>round bars $d/M =$<br>0.29         | 3           | $U/u' = 8.9 + 1.035 x/M$      | 2.20 |

could not be expected to apply to them, and also because, as has been pointed out, there is evidence that the disturbances already in the stream pass through the grids and, when  $M$  is small, become more important than those produced by the grid.

The values of  $A$  calculated by using the relationship

$$\frac{U}{u'} = \text{constant} + \frac{5}{A^2} \frac{x}{M}$$

as identical with the equations in column 4 of Table IV are given in column 5. It will be seen from fig. 6 that though  $U/u'$  ranges from 4.9 to 110, and  $M$  from 0.62 inch to 5 inches all the observed values of  $A$

given in column 5 of Table IV lie in the range from 1.95 to 2.20. This appears to confirm the accuracy of the theoretical prediction made in Part I that  $A$  is a universal constant for turbulence produced or controlled by any type of square grid or honeycomb.

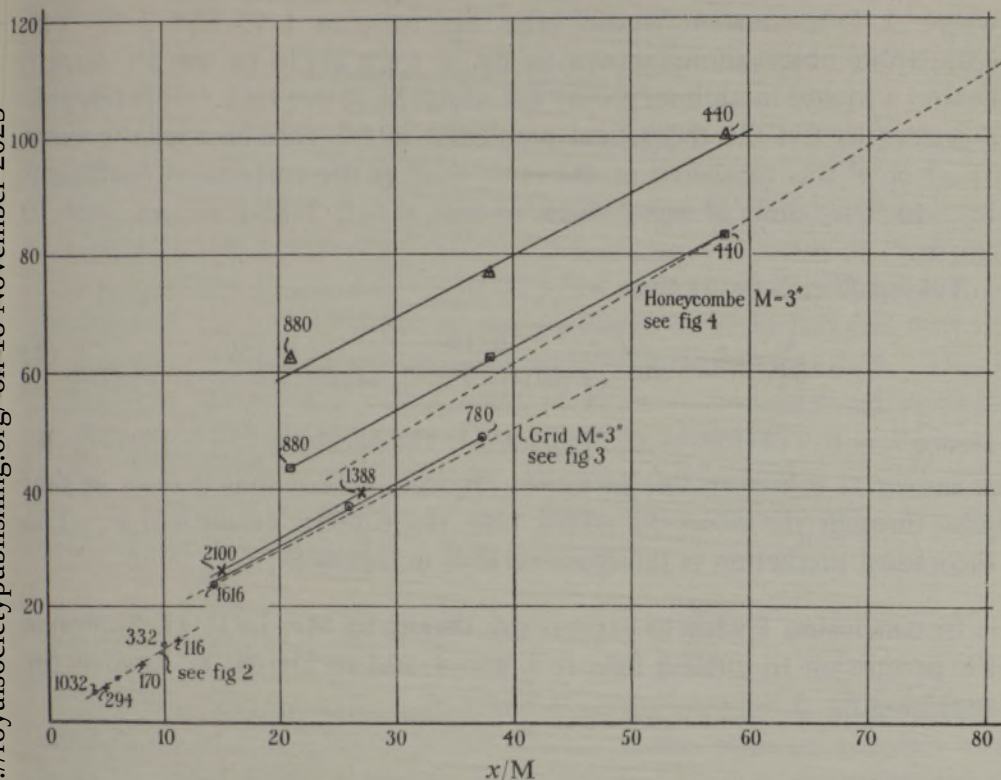


FIG. 6—Collected results of turbulence behind grids and honeycombs.

- |                    |   |                                                                  |
|--------------------|---|------------------------------------------------------------------|
| Dryden             | { | × 5-inch grid, $M = 5$ inches, $d = 1$ inch.                     |
|                    |   | ○ $3\frac{1}{2}$ -inch grid, $M = 3.25$ inches, $d = 0.65$ inch. |
|                    |   | □ Hexagonal honeycomb, $M = 3$ inches.                           |
|                    |   | △ Honeycomb of 3-inch tubes, $M = 3$ inches.                     |
| Simmons and Salter | { | + grid, $M = 0.62$ inch in 4-inch pipe.                          |
|                    |   | • grid, $M = 1.5$ inches in 1-foot tunnel.                       |

### EXPERIMENTAL VERIFICATION OF CONNECTION BETWEEN DISSIPATION OF ENERGY AND INITIAL CURVATURE OF $R_v$ CURVE

In Part I it is predicted that near the origin the correlation curve,  $R_v$ , will coincide with the parabola

$$y = \lambda \sqrt{1 - R_v} \quad (6)$$

where  $\lambda^2 = W/(15 \mu u^2)$  and the dissipation is  $W$ . It has now been shown



that the dissipation behind a honeycomb or grid is in good agreement with observation provided

$$\frac{\lambda}{M} = A \sqrt{\frac{\nu}{u'M}}, \quad (7)$$

where  $A$  is a constant whose value lies between 1.95 and 2.2. The correlation observations shown in fig. 1 were made in the air stream behind a square-mesh honeycomb for which  $M = 0.9$  inch. It is possible therefore to test the theoretical prediction in this case because the value of  $u'^2$  or  $\overline{u'^2}$  was measured at the same time as the correlation coefficient  $R_\nu$ . In ft/sec units  $\overline{u'^2}$  was 0.1015 so that  $u' = 0.3185$  ft per sec = 9.70 cm/sec.

Taking  $A$  as 2.0, (7) gives

$$\frac{\lambda}{M} = 2.0 \sqrt{\frac{0.14}{(9.70)(0.9 \times 2.54)}} = 0.159. \quad (8)$$

Hence  $\lambda = 0.9(0.159) = 0.143$  inches. The parabola  $y = 0.143\sqrt{1-R_\nu}$  is shown as a broken line in fig. 1. It will be seen that it does in fact pass through the observed points near the summit of the curve. The theoretical prediction is therefore verified in this case.

In conclusion I wish to express my thanks to Mr. L. F. G. Simmons for permission to publish figs. 1, 3, and 4, and to Dr. H. L. Dryden for supplying fig. 5.

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